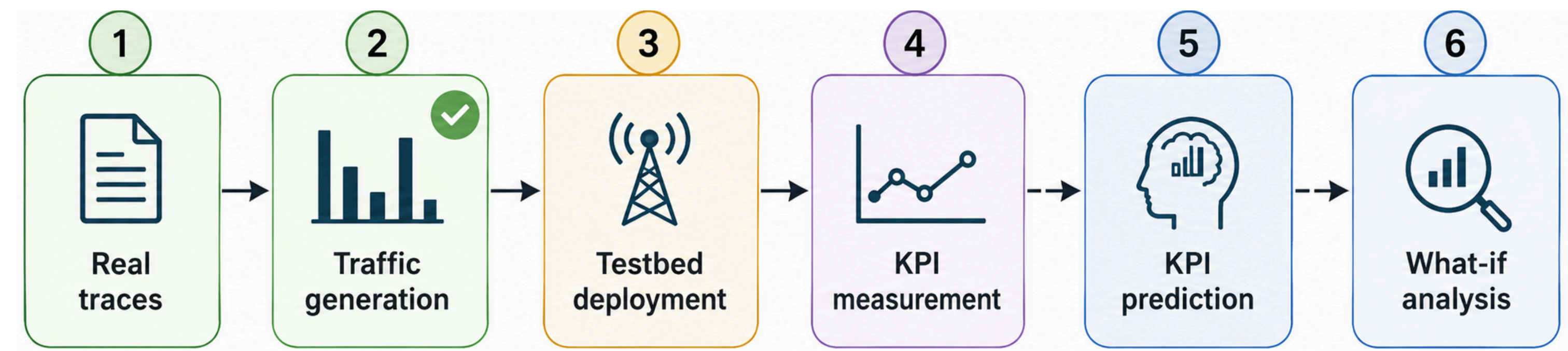


1. Context and Research Gap

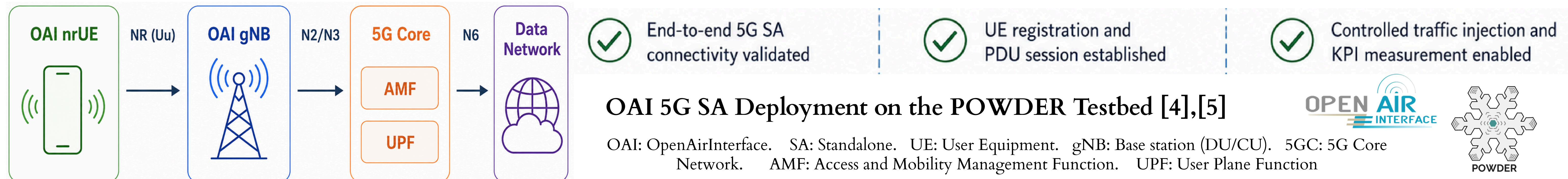
- 5G performance depends on dynamic traffic and radio conditions [1],[2].
- Simplified workloads can produce unreliable KPI evaluations [3].
- Research gap: few existing approaches jointly integrate realistic workloads, reproducible experimentation, and reliable KPI prediction.

Trustworthiness depends on realistic workloads, reproducible measurements, and reliable predictions

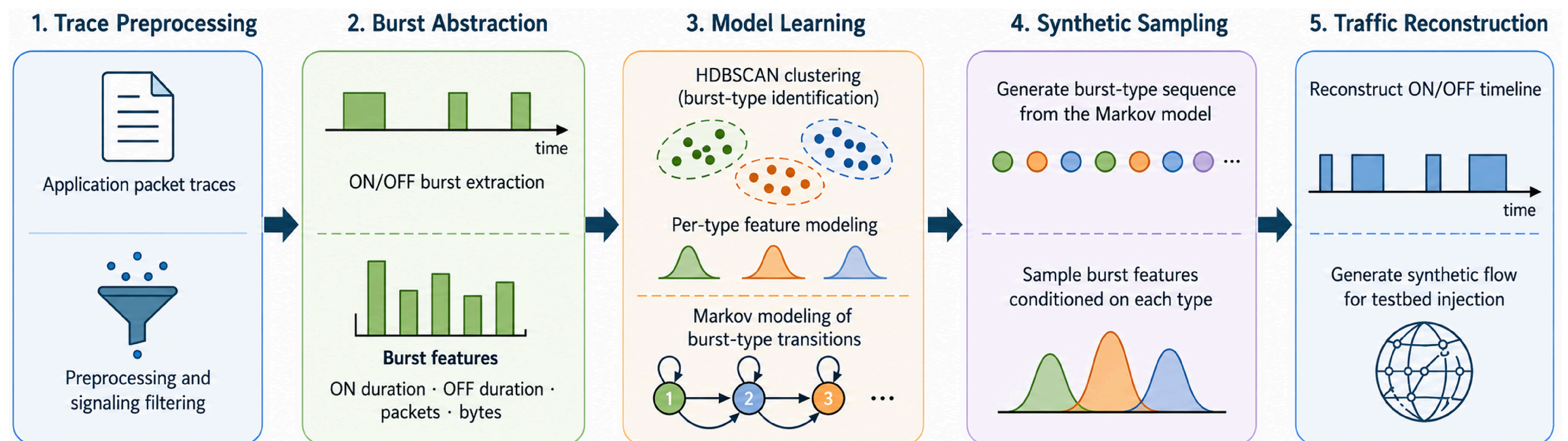
2. Digital Twin Research Framework



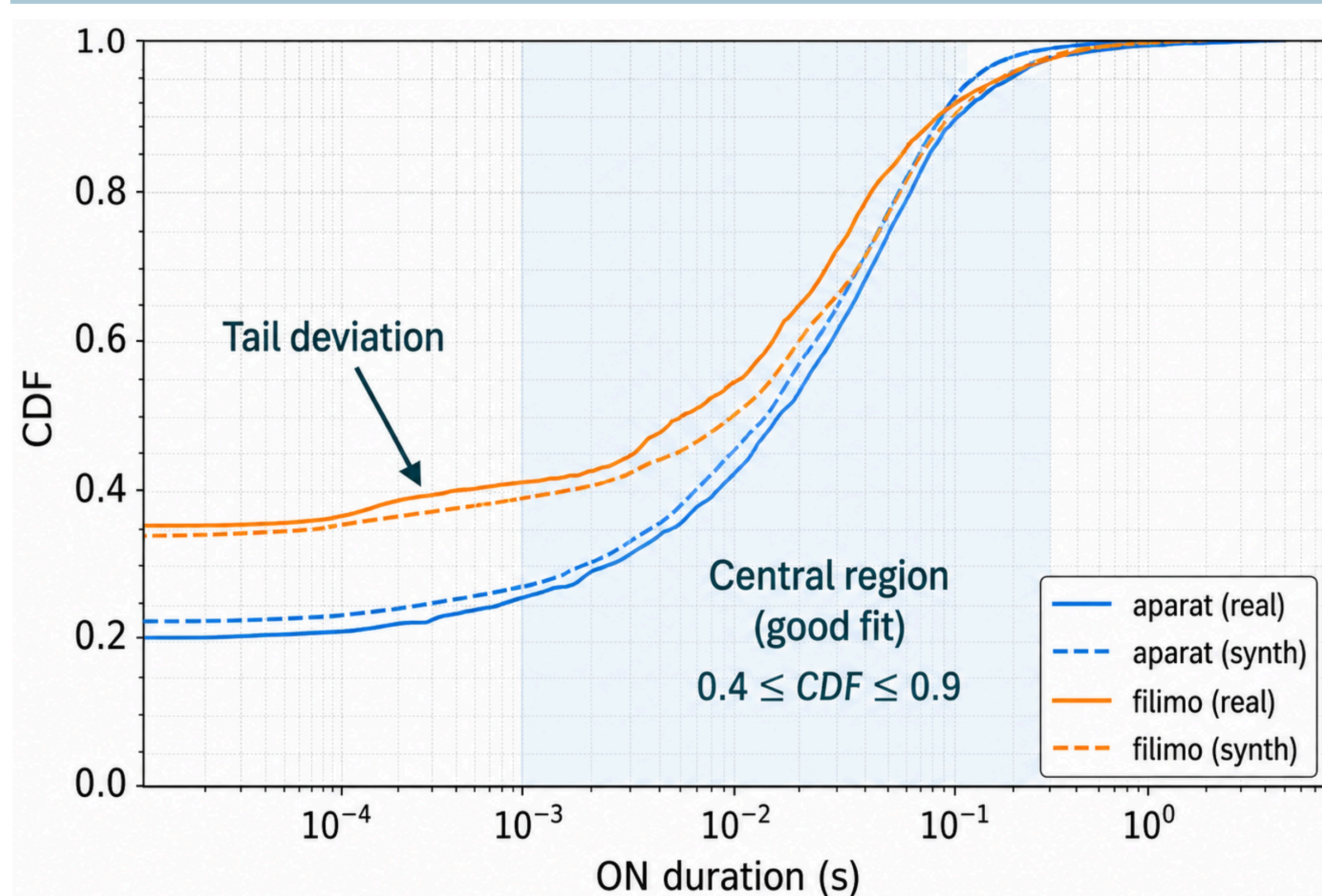
3. Experimental Foundation



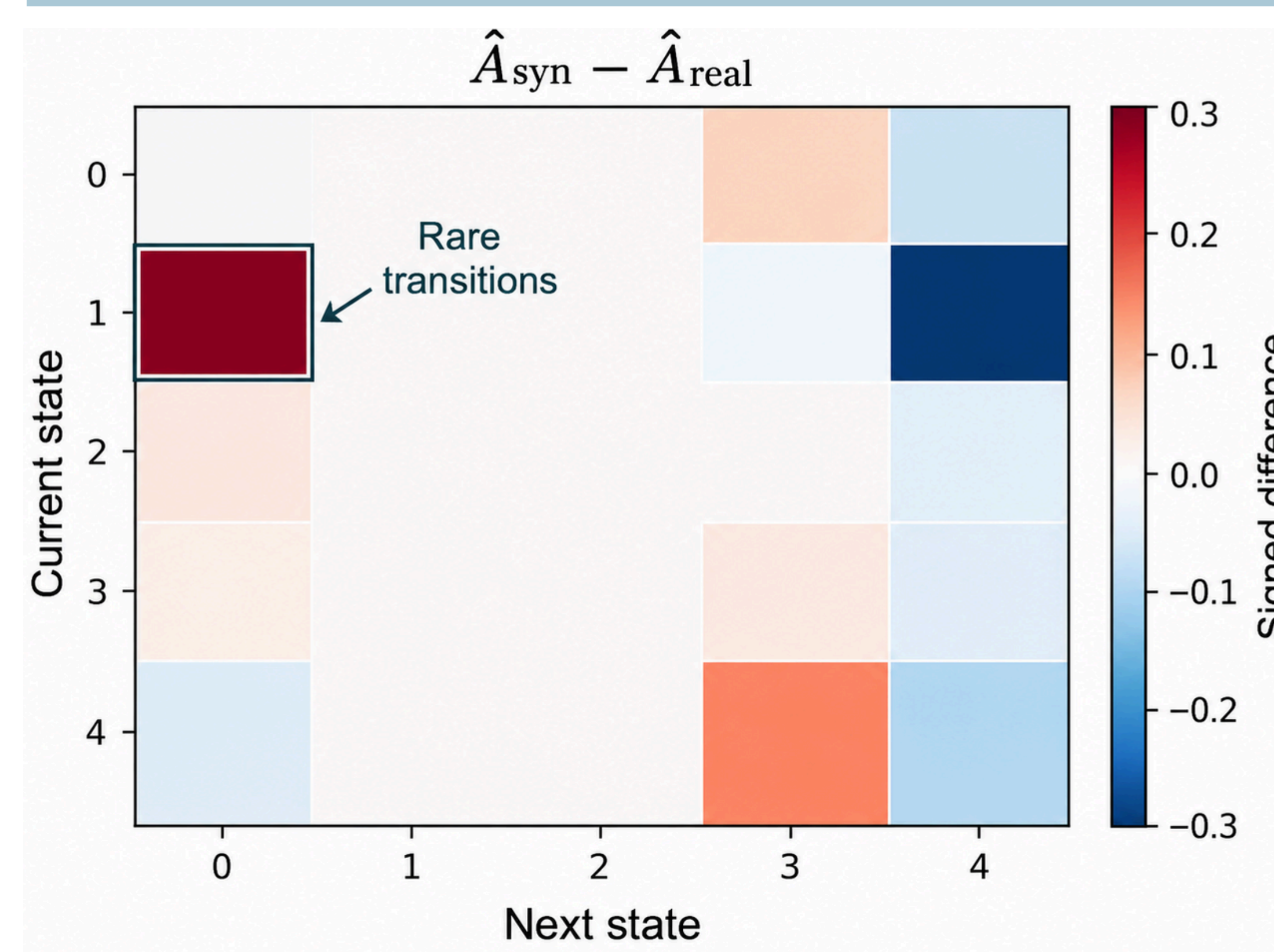
4. Contribution 1: Trace-Driven Generative Traffic Modeling [6]



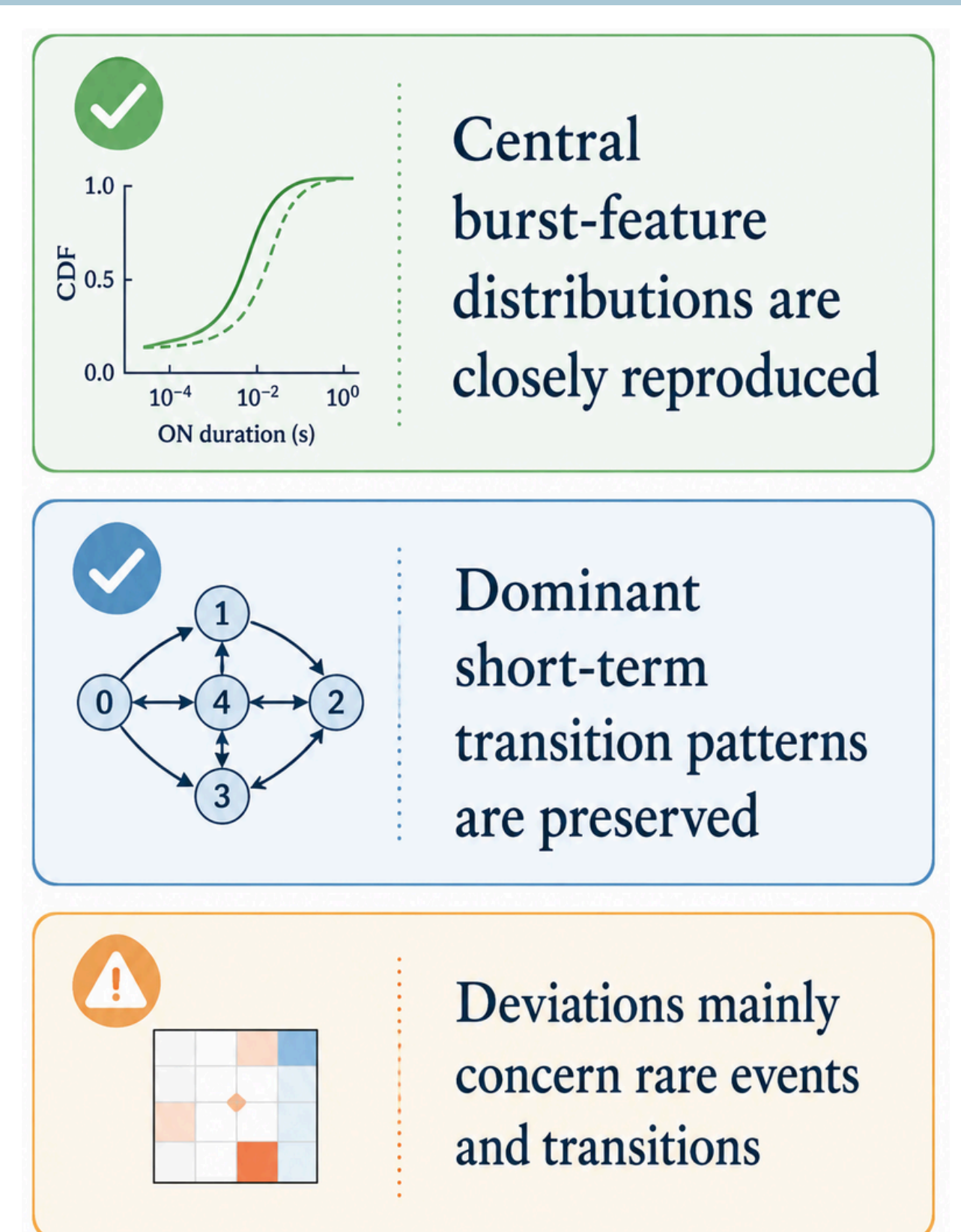
5. Results I: Distributional Fidelity [6]



6. Results II: Temporal Fidelity [6]



Key findings



7. Contribution Summary and Publication

- Models application traffic through an interpretable ON/OFF burst representation.
- Combines application-specific burst types with short-term temporal dependencies.
- Generates lightweight and controllable synthetic workloads.
- Supports repeatable traffic injection in experimental 5G testbeds.

Accepted paper — IEEE NetSoft 2026 [6].

8. Future Work

- Inputs: network state, telemetry, and workload features;
- Prediction task: learn the mapping from network conditions to KPIs;
- Outputs: KPI estimates and prediction intervals;
- Validation: controlled and unseen testbed scenarios;
- Long-term goal: reliable what-if analysis and closed-loop support.

Predictions must be reliable, not only accurate.

References

[1] L. U. Khan et al., "Digital Twin of Wireless Systems: Overview, Taxonomy, Challenges, and Opportunities," IEEE Commun. Surveys & Tutorials, 2022.
 [2] H. X. Nguyen et al., "Digital Twin for 5G and Beyond," IEEE Commun. Mag., 2021.
 [3] L. Bonati et al., "Twinning Commercial Network Traces on Experimental Open RAN Platforms," ACM MobiCom, 2024.

[4] POWDER Wireless, "Platform for Open Wireless Data-driven Experimental Research."
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 [6] G. Ismail, S. Si-Mohammed, and F. Théoleyre, "Multi-Application Generative Traffic Modeling for 5G Digital Twins," IEEE NetSoft, 2026, accepted.